

A Modality-Agnostic Ingestion and Multi-Domain Feature Engineering Pipeline for Heterogeneous Biomedical Data

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Abstract—[CONTEXT: Clinical decision support pipelines for biomedical data are conventionally built around a single modality, so that ingestion and feature extraction must be re-engineered whenever the input modality changes.] [GAP: There is a lack of architectures that infer the structure of an arbitrary biomedical input and adapt reading, preprocessing and feature extraction without manual reconfiguration.] [PROPOSAL: This paper presents the ingestion and feature-engineering stage of BioStatusIA v3, which automatically classifies raw input into one of nine structural modes, validates it, and normalizes every supported modality into a single data contract.] [METHOD: Three biomedical signal families are supported — temporal physiological signals (F1), two-dimensional DICOM imaging (F3) and three-dimensional volumes (F4) — alongside tabular clinical records and plain two-dimensional images. A family-specific reader converts each input into the *SinalNormalizado* contract, from which adaptive preprocessing and multi-domain feature extraction produce a classifier-ready feature vector.] [RESULTS: The pipeline was exercised over thirty real biomedical benchmark datasets covering all families in scope; detected structural mode and feature-vector dimensionality are reported per dataset and per family.] [IMPLICATION: The results indicate that structural detection operates on the physical organization of the data rather than on its declared clinical semantics, which is discussed as both a strength and a limitation.]

Index Terms—biomedical signal processing, data ingestion, feature engineering, radiomics, multimodal data, clinical decision support systems, DICOM, medical imaging

I. INTRODUCTION

Clinical decision support systems (CDSS) are increasingly expected to operate over the full range of data a healthcare institution produces: physiological recordings acquired at the bedside, imaging studies stored in a picture archiving system, volumetric acquisitions from computed tomography or magnetic resonance scanners, and structured clinical variables held in electronic records. Research attention, however, concentrates almost entirely on the modeling stage — which classifier is used, how it is validated, how its output is explained. The stage that precedes modeling receives far less scrutiny, even though it determines what the model is ever able to see. Before any classifier is trained, raw input must be recognized, read,

cleaned and converted into a numerical representation, and every design decision taken at that point silently constrains everything downstream.

A. Context and Problem Statement

In practice, biomedical pipelines are mono-modal. Ingestion and feature extraction are written against one modality — electrocardiography, or chest radiography, or a tabular registry — and the reading routines, the preprocessing operations and the descriptor set are all specialized to that choice. The consequence is structural rather than incidental: when the modality changes, the pipeline does not adapt, it is rewritten. An institution holding physiological signals, two-dimensional imaging studies, volumetric acquisitions and tabular records therefore pays the engineering cost once per family of data, and obtains no representation that is shared across them.^[cite: survey on multimodal CDSS pipelines]

Two consequences follow. First, feature engineering cannot be reused: a texture descriptor implemented for mammography and a spectral descriptor implemented for electroencephalography live in disjoint code paths, even though both ultimately produce a numeric vector for the same downstream consumer. Second, supporting a new modality becomes a code change rather than a configuration change, because the modality is implicitly encoded in the control flow of the pipeline instead of being an explicit property of the data.

This work addresses the stage that precedes modeling, and asks the following research question:

How can a biomedical pipeline infer the structure of an arbitrary input and adapt reading, preprocessing and feature extraction to it, without manual reconfiguration and without modality-specific branching leaking into the extraction stage?

The system studied here, BioStatusIA, addresses this question with four coupled mechanisms: a deterministic classification of the incoming input into one of nine structural modes; a validation layer that confirms readability by the reference library of the corresponding family before any heavy processing begins; a single data contract into which every

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supported modality is normalized; and a preprocessing stage whose operations are selected from measured properties of the data rather than from a fixed per-modality configuration. Table I summarizes the data families within the scope of this work.

Table I
BIOMEDICAL DATA FAMILIES WITHIN THE SCOPE OF THIS WORK

Family	Types	Reading libraries
F1	ECG, EEG, EMG, EOG, PPG, spirometry	mne, wfdb, scipy
F3	Radiography, mammography, ultrasound	pydicom
F4	CT, MRI, PET/SPECT	nibabel, SimpleITK
Tabular	Clinical variables	custom parser
2D image	Non-DICOM images	OpenCV, scikit-image

B. Objectives

The general objective of this work is to design, implement and empirically evaluate a modality-agnostic ingestion and feature-engineering pipeline that automatically detects the structure of heterogeneous biomedical input, normalizes it into a single data contract, and produces a classifier-ready multi-domain feature vector without manual reconfiguration.

The specific objectives are:

- to formalize an automatic input-structure detection mechanism covering nine input modes, and to couple it with schema-level validation that confirms readability before processing;
- to define a unified data contract that decouples family-specific readers from family-specific extractors, and to verify that it serves temporal signals, two-dimensional DICOM studies, volumetric acquisitions, tabular records and plain images without modification;
- to specify an adaptive preprocessing stage whose operations are selected from measurable properties of the input and whose decisions are individually justified;
- to characterize the multi-domain descriptor set produced for each family, spanning the time domain, the frequency domain, texture, morphology and volumetric radiomics; and
- to evaluate ingestion robustness and feature coverage over thirty real biomedical benchmark datasets spanning every family in scope.

C. Contributions

The contributions of this paper are the following.

- 1) A deterministic, rule-based mechanism that classifies an arbitrary file, archive or directory into one of nine structural modes, combining extension predicates, directory inventory, recursive label-directory search and cardinality thresholds (Section III-B).

- 2) A unified data contract, *SinalNormalizado*, which decouples readers from extractors so that supporting an additional family requires one reader and one extractor and no change to the orchestration layer (Section III-D).
- 3) An adaptive preprocessing stage in which each selected operation is triggered by a measured property of the dataset and is accompanied by a machine-generated justification recording the value that triggered it (Section III-F).
- 4) A multi-domain handcrafted descriptor set covering time-domain and spectral descriptors for physiological signals, texture and morphology for two-dimensional imaging, and volumetric radiomics for three-dimensional acquisitions (Section III-G).
- 5) An empirical evaluation of ingestion robustness and feature coverage over thirty real biomedical datasets, reporting the detected structural mode and the resulting feature-vector dimensionality for each of them (Section IV).

D. Scope Boundary

This paper deliberately terminates at the feature vector. Classifier training, model selection, probability calibration, interpretability analysis and automated report generation are addressed in a companion work, and no predictive performance is reported here. The boundary is stated explicitly because the contribution under evaluation is the ingestion and representation stage: mixing it with downstream classification results would obscure which properties of the system are being assessed.

E. Paper Organization

The remainder of this paper is organized as follows: Section II provides background information and related work; Section III outlines the proposed ingestion and feature-engineering architecture; Section IV presents and discusses the experimental evaluation; and Section V concludes the paper.

II. BACKGROUND AND RELATED WORK

This section establishes the material the proposed pipeline operates on and the analytical tools it applies. Section II-A characterizes the biomedical data families by their physical encoding, which is the property the detection mechanism actually observes. Sections II-B and II-C review the descriptor families used for imaging and for physiological signals, respectively. Section II-D surveys ingestion architectures and data contracts, and positions the present contribution against existing work.

A. Biomedical Data Families and Their Encodings

A recurring source of confusion in multimodal pipelines is the conflation of the *clinical semantics* of a recording with its *physical encoding*. A twelve-lead electrocardiogram is semantically a temporal physiological signal regardless of whether it is distributed as a WFDB record, as a European Data Format recording, or as a comma-separated table of sampled amplitudes. The encoding, not the semantics, determines

which library can read the file and which descriptors can be computed from it. This work therefore organizes the input space by encoding.

Family F1 comprises temporal physiological signals: electrocardiography, electroencephalography, electromyography, electro-oculography, photoplethysmography, blood pressure and spirometry. Its encodings include WFDB record and header pairs, European Data Format recordings and their extended biosemi variant, MATLAB arrays, vendor-specific spirometry documents in XML, and motion-capture and polysomnography containers. ^{[cite: PhysioNet / WFDB][cite: EDF standard]} Family F3 comprises two-dimensional imaging studies encoded as single DICOM instances, in which pixel data is accompanied by an acquisition header carrying modality, spatial calibration and display parameters. ^[cite: DICOM standard] Family F4 comprises volumetric acquisitions — computed tomography, magnetic resonance and positron emission tomography — encoded as NIfTI or MetaImage volumes, or reconstructed from a directory holding a DICOM series. ^[cite: NIfTI format] Tabular clinical records are encoded as delimited text, and plain two-dimensional images as conventional raster formats without an acquisition header.

The distinction matters for a further reason: the amount of contextual metadata differs sharply across encodings. A DICOM instance carries its own modality declaration and spatial calibration; a NIfTI volume carries voxel spacing but little acquisition context; a delimited text file carries nothing beyond column names. Any modality-agnostic pipeline must therefore tolerate a wide range of metadata availability rather than assuming a uniform descriptive header.

B. Radiomics and Handcrafted Feature Extraction

Radiomics denotes the extraction of large numbers of quantitative descriptors from medical images, under the hypothesis that image texture and shape encode clinically relevant information not apparent to visual inspection. ^[cite: radiomics foundational work] A substantial body of work has standardized the definition and computation of such descriptors so that values obtained by different implementations remain comparable. ^[cite: image biomarker standardization]

Two broad strategies exist for producing a numerical representation of an image. Learned representations, typically obtained from convolutional networks, require a training stage and large labeled corpora, and yield features whose individual meaning is not directly interpretable. Handcrafted descriptors are computed analytically from explicit definitions, require no training, and remain interpretable in clinical terms. The present work adopts handcrafted descriptors at the ingestion stage for three reasons: they impose no training dependency on a stage that must run before any model exists; they behave predictably in the small-sample regimes common in clinical datasets; and their interpretability is a prerequisite for the auditability the system targets. A learned-representation branch is retained as an extension point but is not evaluated here.

Second-order texture is characterized through the gray-level co-occurrence matrix (GLCM). ^[cite: Haralick texture features] Let $p(i, j)$ denote the normalized probability of observing gray levels i and j at a given offset. Entropy, which increases with tissue heterogeneity, is defined as

$$H = - \sum_{i,j} p(i, j) \log p(i, j), \quad (1)$$

and contrast, which increases with local intensity variation, as

$$C = \sum_{i,j} (i - j)^2 p(i, j). \quad (2)$$

Homogeneity, $\sum_{i,j} p(i, j)/(1 + |i - j|)$, and energy, $\sum_{i,j} p(i, j)^2$, complete the texture set adopted in this work.

Morphological regularity is quantified by circularity, $4\pi A/P^2$, and by solidity, A/A_{hull} , where A and P denote the area and perimeter of the segmented region and A_{hull} the area of its convex hull. Both decrease as contours become irregular. For volumetric data the analogous descriptor is sphericity,

$$\Psi = \frac{\pi^{1/3} (6V)^{2/3}}{S}, \quad (3)$$

with V and S the volume and surface area of the candidate lesion; Ψ attains unity for a perfect sphere and decreases with morphological complexity.

C. Physiological Signal Processing

Descriptors for family F1 are drawn from two complementary domains. In the time domain, signal energy is summarized by the root mean square,

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}, \quad (4)$$

complemented by dispersion, skewness, kurtosis, peak-to-peak amplitude and a signal-to-noise ratio expressed in decibels. In the frequency domain, the power spectral density is estimated by averaging periodograms over overlapping windowed segments, a procedure that trades spectral resolution for variance reduction and is therefore well suited to non-stationary physiological recordings. ^[cite: Welch periodogram averaging] From the resulting spectrum the pipeline derives total power, the spectral centroid, the dominant frequency, the median frequency and the power contained in fixed frequency bands.

Modality-specific descriptors are layered on top of this common basis. For electroencephalography, relative band power is computed as

$$P_b = \int_{f_1}^{f_2} S(f) df \bigg/ \int_{f_{\min}}^{f_{\max}} S(f) df, \quad (5)$$

over the delta, theta, alpha, beta and gamma bands. For electrocardiography, R-peak detection yields the sequence of interbeat intervals from which mean heart rate and heart-rate

variability are derived. Variability is characterized by the root mean square of successive differences,

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}, \quad (6)$$

by the standard deviation of the intervals, and by the proportion of successive intervals differing by more than fifty milliseconds.^[cite: HRV measurement standards] For electromyography, the envelope root mean square and the median frequency are used. For spirometry, the pipeline recovers forced vital capacity, forced expiratory volume in one second, their ratio, and peak expiratory flow.

D. Ingestion Architectures and Data Contracts

Interoperability standards define how biomedical data is exchanged, and are mature for both imaging and clinical records.^{[cite: DICOM standard][cite: HL7 FHIR]} They specify encoding and transport, however, and not the analytical representation a machine-learning pipeline consumes. The translation from an interoperable encoding to a feature vector remains an application concern, and is where per-modality specialization typically accumulates.

Within machine-learning engineering, the notion of a *data contract* — an explicit, validated interface between the producer and the consumer of a dataset — has been advanced as a way to prevent silent schema drift from propagating into models.^[cite: data contracts / schema validation in ML systems] Related work applies declarative schema validation at pipeline boundaries so that malformed input fails early and diagnosably rather than deep inside a numerical routine.^[cite: data validation in production ML]

Three lines of existing work are adjacent to the present contribution. Single-modality clinical pipelines achieve high specialization but treat the modality as a compile-time constant. General-purpose radiomics toolkits offer standardized and extensively validated descriptor implementations, but assume that the modality and the region of interest are known before extraction begins, and therefore address the stage *after* the one studied here.^[cite: radiomics toolkit] End-to-end deep pipelines bypass explicit feature engineering altogether, at the cost of the interpretability and the training-free operation that the ingestion stage requires.

What is missing across these lines is a mechanism that treats the structure of the input as something to be *inferred* rather than declared, and that converts every inferred family into a single representation before extraction begins. The architecture described in Section III occupies precisely that gap.

III. METHODOLOGY

This section describes the architecture of the proposed ingestion and feature-engineering stage and the protocol used to evaluate it. The presentation follows the order in which data traverses the system: structural detection, schema validation, family-specific reading, normalization into the unified contract, adaptive preprocessing, multi-domain extraction and vector consolidation. The evaluation protocol is stated last,

and is deliberately restricted to properties of ingestion and representation.

A. Pipeline Overview

Figure 1 shows the complete path. An arbitrary input — a single file, a compressed archive or a directory tree — is first classified into one of nine structural modes by a deterministic rule set. The inferred mode selects a validation schema, which confirms that the input is readable by the reference library of the corresponding family and extracts a metadata card from its header. Only then is the file read in full, by a family-specific reader that converts it into the *SinalNormalizado* contract. From that point onward no component observes the original file format: preprocessing and extraction operate exclusively on the contract, and the stage terminates by emitting a numerical feature vector.

The architectural consequence of this ordering is that modality-specific knowledge is confined to two layers — the detection rules and the readers — and never reaches the orchestration layer.

B. Automatic Input-Structure Detection

Detection is deterministic and rule-based rather than learned, so that its behaviour is fully auditable and reproducible. Two cases are distinguished.

When the input is a single file, the decision reduces to an ordered sequence of extension predicates: raster image formats yield the single-image mode; delimited text yields the tabular mode; temporal-signal containers yield the temporal-signal mode; a single DICOM instance yields the two-dimensional DICOM mode; and NIfTI or MetaImage files yield the volumetric mode. An input matching no predicate is rejected as invalid rather than processed on a best-effort basis. Mask files are excluded from the image predicate by name, so that segmentation annotations distributed alongside a dataset are not mistaken for samples.

When the input is a directory, the tree is inventoried recursively and reduced to five boolean indicators recording whether tabular files, plain images, temporal signals, DICOM instances or volumetric files are present. The decision then proceeds by the following rules. If more than one indicator is set, and the combination is not the specific pair of tabular files and plain images, the input is classified as expanded multimodal. That pair is treated separately because it corresponds to the common case of an image dataset accompanied by a clinical table, which the system handles as a distinct multimodal mode. If only tabular files are present, the tabular mode is selected. Otherwise the tree is searched recursively for label directories; when both a negative and a positive label directory are found, the input is classified as a labeled dataset, irrespective of how deeply the labels are nested. Failing that, the remaining single-family cases are resolved directly, with one cardinality rule: a directory containing at least ten DICOM instances is interpreted as a volumetric series, whereas a smaller number is interpreted as a set of independent two-dimensional studies. Table II summarizes the nine resulting modes.

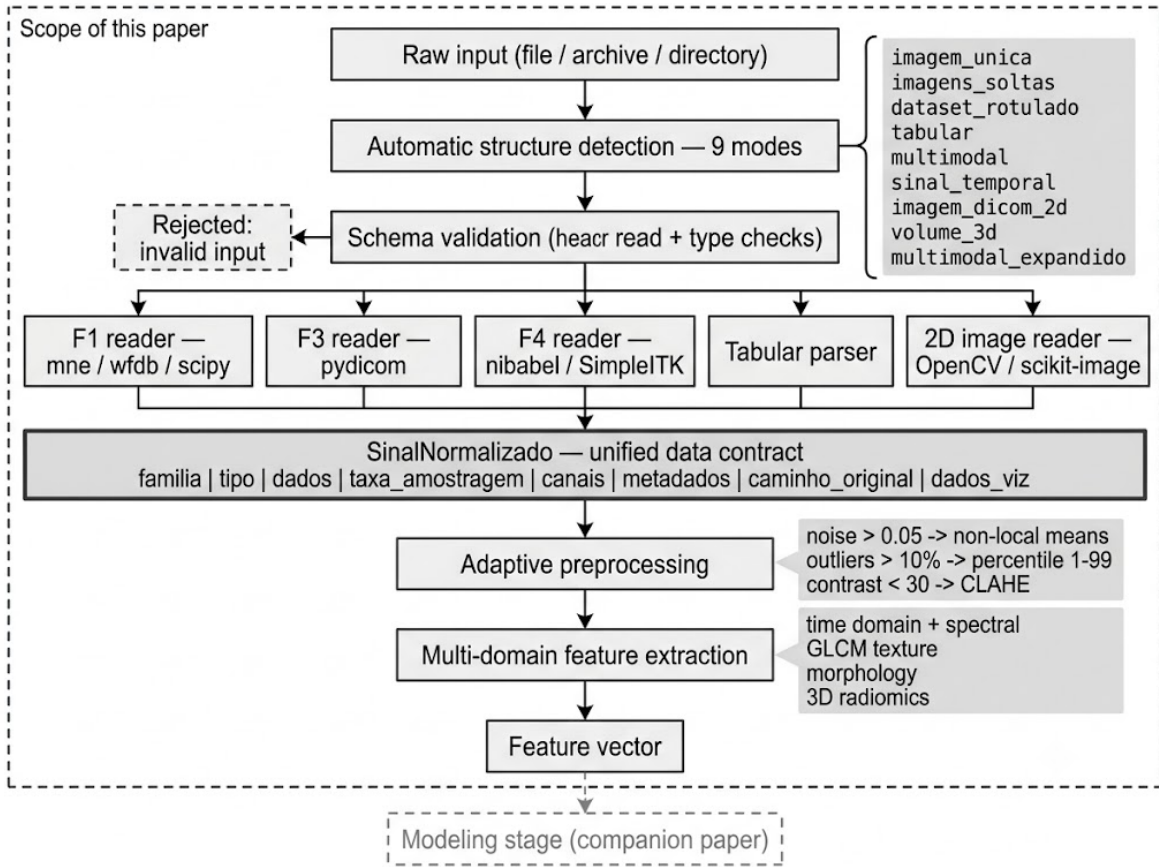


Figure 1. Block diagram of the proposed ingestion and feature-engineering pipeline. Raw input is classified into one of nine structural modes, validated against a family schema, and read by a family-specific reader into the *SinalNormalizado* contract, upon which adaptive preprocessing and multi-domain extraction operate. The dashed boundary marks the scope of this paper; the modeling stage is addressed in a companion work.

Table II
AUTOMATICALLY DETECTED INPUT MODES

Mode	Detection criterion	Family
imagem_unica	single 2D image file	2D image
imagens_soltas	images, no label directories	2D image
dataset_rotulado	label subdirectories present	2D image
tabular	CSV/TXT/TSV only	Tabular
multimodal	images and tabular together	mixed
sinal_temporal	WFDB, EDF, MAT, XML or C3D	F1
imagem_dicom_2d	single DICOM instance	F3
volume_3d	NIfTI/MHA or ≥ 10 DICOM	F4
multimodal_expandido	images, signals and tabular	mixed

Label recognition relies on a fixed vocabulary of directory names, with one set denoting the negative class and another the positive class. The vocabulary is configuration rather than control flow: extending it to a new naming convention does not alter the detection procedure.

C. Schema Validation

Detection by extension is a necessary but insufficient criterion, since an extension asserts an intention rather than a fact. A validation layer therefore intervenes between detection and reading. For each family, the layer attempts an actual header read using the reference library of that family and populates a declarative metadata card containing the modality, the sampling rate, the number and names of channels, and the array dimensions, each field constrained by an explicit type and range. Failures are captured as human-readable messages attached to a validated-input record, whose validity flag gates all subsequent processing.

The purpose of this layer is diagnostic rather than defensive. A file whose extension declares a family it does not belong to fails at the header, before any expensive decoding is attempted, and reports which field or which library raised the failure — as opposed to surfacing an opaque numerical error deep inside an extractor.

For tabular input, validation additionally infers the schema. The label column is identified by name against a vocabulary of common clinical designations; when no name matches, the procedure falls back to the last column presenting between two and ten distinct values. The field delimiter is inferred among

comma, semicolon, tabulation and vertical bar, and character encoding is attempted as UTF-8 with a Latin-1 fallback.

D. Unified Data Contract

The central architectural decision of this work is that every reader emits, and every extractor consumes, a single structure. Table III lists its fields.

Table III
FIELDS OF THE SINALNORMALIZADO DATA CONTRACT

Field	Type	Role
familia	str	family identifier
tipo	str	modality label
dados	ndarray	raw array
taxa_amostragem	float	sampling rate (Hz)
canais	list	channel labels
metadados	dict	source header
caminho_original	str	provenance
dados_viz	list	downsampled series

The array field carries a family-dependent shape: channels by samples for temporal signals, height by width for two-dimensional images, and depth by height by width for volumes, the latter two normalized to single-precision values in the unit interval. The sampling-rate field is set to zero for families in which it does not apply, rather than omitted, so that the contract remains structurally identical across families. The visualization field holds a downsampled copy limited to two thousand points per channel, which keeps the interface responsive without the interface layer ever touching the analytical array.

The invariant is strict: readers and extractors never reference one another. The practical consequence, and the property this work claims, is that supporting an additional family requires one new reader and one new extractor, with no modification to detection, orchestration or consolidation.

E. Family-Specific Readers

A single dispatch function maps the detected family to its reader. For family F1, WFDB record and header pairs, European Data Format recordings, MATLAB arrays, spirometry documents in XML and motion-capture containers are each read by the corresponding reference library, from which the sampling rate and channel labels are recovered directly; the modality label is inferred from the container type.^{[cite: MNE-Python][cite: PhysioNet / WFDB]}

For family F3, the DICOM instance is parsed and the display window is applied automatically from the window-center and window-width header tags, so that the intensity range presented to the extractor reflects the acquisition rather than an arbitrary rescaling; the modality is recovered from the header.^[cite: pydicom]

For family F4, NIFTI volumes are read and transposed from the native axis ordering to an axial-first convention, so that the first array axis is the slice axis for every volumetric input regardless of source format; MetaImage volumes are read through a second library, from which voxel spacing in

millimetres is recovered and retained in the metadata, since it is required to express volumetric descriptors in physical units.^{[cite: nibabel][cite: SimpleITK]}

F. Adaptive Preprocessing

Preprocessing is decided rather than configured. A base-analysis step first measures properties of the incoming dataset — mean noise level, the number of samples presenting outlying intensity distributions, and mean contrast. A strategy-decision step then selects operations from those measurements according to explicit thresholds, and records for each selected operation a machine-generated justification containing the measured value that triggered it.

Three decisions are taken. When mean noise exceeds 0.05, non-local means denoising is selected; otherwise a 5×5 Gaussian blur is applied. When the number of intensity outliers exceeds ten percent of the dataset, normalization is performed against the first and ninety-ninth percentiles; otherwise min-max normalization is used. When mean contrast falls below 30, contrast-limited adaptive histogram equalization is applied. Resizing to 256×256 is unconditional.

Two properties of this design are worth stating. First, the selection is a function of the dataset rather than of the modality, so two datasets of the same modality may legitimately receive different treatment. Second, because every decision carries the value that produced it, the preprocessing stage is auditable after the fact — which is a requirement in a clinical support context, where the provenance of a transformation may need to be reconstructed.

G. Multi-Domain Feature Extraction

Extraction is dispatched by family and produces named descriptors.

For family F1, the common basis comprises the root mean square, the mean, the standard deviation, skewness, kurtosis, peak-to-peak and maximum amplitude, and the signal-to-noise ratio in decibels, together with spectral descriptors: total power, spectral centroid, dominant and median frequency, and power in four fixed bands. Modality-specific descriptors are layered on top. Electroencephalographic recordings additionally yield relative power in the delta, theta, alpha, beta and gamma bands. Electrocardiographic recordings yield the number of detected R-peaks, mean heart rate, the mean and standard deviation of interbeat intervals, and the three variability descriptors defined in Section II-C. Electromyographic recordings yield the envelope root mean square, and spirometry documents yield forced vital capacity, forced expiratory volume in one second, their ratio, and peak expiratory flow.

For family F3, the descriptor set combines morphology (circularity and solidity), GLCM texture (entropy, homogeneity, energy and contrast), intensity distribution (signal-to-noise ratio, skewness and kurtosis) and DICOM-specific descriptors: the fraction of high-density pixels, the mean gradient, uniformity, the declared modality and the pixel spacing.

For family F4, global volumetric statistics comprise the mean, standard deviation and median intensity, the fifth and

ninety-fifth percentiles, and the count and fraction of high-intensity voxels. Texture is computed as GLCM descriptors on each of the three orthogonal planes — axial, coronal and sagittal — rather than on a single representative slice. Volumetric morphology yields the candidate lesion volume expressed in cubic millimetres using the recovered voxel spacing, and an approximate sphericity.

For tabular input, the descriptors are the native clinical variables of the source dataset after schema inference. For plain two-dimensional images, the descriptor set is the morphology, texture and intensity groups described above, without the DICOM-specific additions.

H. Feature Vector Consolidation

Consolidation flattens the named descriptor groups into a single numerical vector, preserving descriptor names so that each position remains traceable to its definition. Descriptors that cannot be computed for a given sample — for instance, R-peak variability on a recording in which no peak is detected — are recorded as absent rather than imputed with a neutral value, so that absence is distinguishable from a measured zero downstream.

The resulting vector is the artifact this stage delivers. Its consumption by the modeling stage lies outside the scope of this paper.

I. Experimental Environment

The pipeline is implemented in Python 3.11. [cite: library versions and hardware specification pending measurement]

J. Evaluation Protocol

The evaluation exercises the pipeline over thirty real biomedical benchmark datasets, six for each of the five families in scope: temporal physiological signals, two-dimensional DICOM studies, volumetric acquisitions, tabular clinical records and plain two-dimensional imaging. [cite: source publication of each of the thirty benchmark datasets]

Four quantities are recorded, and only these four. First, the structural mode assigned to each dataset by the detection procedure. Second, whether ingestion completed without unhandled exceptions. Third, the dimensionality of the resulting feature vector. Fourth, which descriptors were effectively produced for each family.

No predictive performance is measured in this paper. The datasets are used as ingestion stimuli rather than as classification tasks, and the question under evaluation is whether a single pipeline can consume heterogeneous real-world inputs and emit a well-formed representation for each of them without per-dataset reconfiguration.

IV. RESULTS AND DISCUSSION

This section reports the behaviour of the ingestion and feature-engineering stage over the benchmark. Four quantities are presented, in the order defined by the evaluation protocol: the structural mode assigned to each dataset, whether ingestion completed without unhandled exceptions, the dimensionality

of the resulting feature vector, and the descriptors effectively produced for each family. No predictive performance is reported, for the reasons stated in Section I-D.

A. Benchmark Composition

The pipeline was exercised over thirty real biomedical datasets, organized as six datasets for each of the five families in scope. Family F1 is represented by electrocardiographic, encephalographic, electromyographic, polysomnographic and spirometric recordings; family F3 by chest radiography and mammography collections; family F4 by brain, lung, liver and prostate volumetric acquisitions obtained by magnetic resonance, computed tomography and positron emission tomography; the tabular family by clinical registries covering cardiac, metabolic, renal, neurological and oncological outcomes; and the plain two-dimensional imaging family by ultrasound, dermatological, histopathological, retinal, haematological and radiographic collections. [cite: source publication of each of the thirty benchmark datasets]

The datasets were selected for encoding heterogeneity rather than for clinical comparability. They differ in file format, in directory organization, in the presence or absence of label directories, in metadata richness and in the physical nature of the recorded quantity. This is the property under test: the benchmark is intended to stress the ingestion mechanism, not to constitute a set of comparable classification tasks.

B. Detection Outcome and Feature Dimensionality

Table IV reports, for each dataset, the structural mode assigned by the detection procedure and the dimensionality of the resulting feature vector. All thirty inputs were processed to completion without unhandled exceptions, and every input produced a well-formed feature vector.

Table V aggregates the detection outcome by family. Agreement is counted at the family level: a detection is considered concordant when the assigned mode belongs to the family the dataset actually represents. Twenty-eight of the thirty datasets, or approximately 93%, were assigned to a mode of the correct family.

Two observations follow, and both concern the nature of the detection mechanism rather than defects in its implementation.

The two divergent cases are datasets 04 and 06, which are semantically electromyographic and spirometric recordings — that is, family F1 — but which are distributed as delimited text files rather than in a native signal container. The detection procedure assigned them to the tabular mode, which is the correct decision given the encoding it observes, and an incorrect one given the clinical semantics a human reader would attribute to them. This is the central qualitative finding of the evaluation: *structural detection operates on the physical organization of the data, not on its declared clinical semantics*. The consequence is not merely taxonomic. A recording routed to the tabular mode is described by its column schema rather than by spectral and variability descriptors, so the divergence propagates into the representation.

Table IV
DETECTED STRUCTURAL MODE AND FEATURE-VECTOR DIMENSIONALITY ACROSS THE BENCHMARK

#	Dataset	Family	Detected mode	Feats.	#	Dataset	Family	Detected mode	Feats.
01	PTB-XL ECG	F1	signal_temporal	18	16	IXI Brain MRI	F4	volume_3d	22
02	MIT-BIH Arrhythmia	F1	signal_temporal	18	17	OASIS-3 Brain PET	F4	volume_3d	22
03	CHB-MIT EEG	F1	signal_temporal	18	18	ProstateX MRI	F4	volume_3d	22
04	EMG Physical Action	F1	tabular	18	19	UCI Heart Disease	Tab.	tabular	6
05	CAP Sleep PSG	F1	signal_temporal	18	20	Breast Cancer Wisconsin	Tab.	tabular	30
06	Spirometry COPD	F1	tabular	18	21	PIMA Diabetes	Tab.	tabular	6
07	NIH ChestXray14	F3	image_dicom_2d	14	22	Parkinson's Biomarkers	Tab.	tabular	16
08	CBIS-DDSM Mammography	F3	image_dicom_2d	14	23	Chronic Kidney Disease	Tab.	tabular	5
09	CheXpert Chest	F3	image_dicom_2d	14	24	Stroke Prediction	Tab.	tabular	5
10	RSNA Pneumonia	F3	image_dicom_2d	14	25	BUSI Breast Ultrasound	2D	dataset_rotulado	12
11	INbreast Mammography	F3	image_dicom_2d	14	26	HAM10000 Dermatology	2D	dataset_rotulado	12
12	VinDr-CXR	F3	image_dicom_2d	14	27	BreakHis Histopathology	2D	dataset_rotulado	12
13	BraTS Brain MRI	F4	volume_3d	22	28	DRIVE Retinal Fundus	2D	images_soltas	12
14	LIDC-IDRI Lung CT	F4	volume_3d	22	29	BCCD Blood Cells	2D	dataset_rotulado	12
15	LiTS Liver CT	F4	volume_3d	22	30	COVID-19 Radiography	2D	dataset_rotulado	12

Table V
DETECTION AGREEMENT BY FAMILY

Family	Datasets	Concordant	Divergent cases
F1	6	4	datasets 04 and 06
F3	6	6	—
F4	6	6	—
Tabular	6	6	—
2D image	6	6	—
Total	30	28	—

Dataset 28 requires a separate remark, because it may appear divergent and is not. It was assigned to the loose-images mode rather than to the labeled-dataset mode, since its directory tree contains no label directories. Both modes belong to the plain two-dimensional imaging family, and the assignment reflects a genuine property of the input: the dataset carries no directory-level class partition for the detector to find. The case is reported here because it illustrates that the detector discriminates not only between families but also between organizational variants within a family.

C. Feature Coverage per Family

Table VI summarizes the dimensionality of the emitted feature vector by family. The result is sharply structured: within each of the four signal-bearing families, dimensionality is constant across all six datasets, and varies only for tabular input.

Table VI
FEATURE-VECTOR DIMENSIONALITY PER FAMILY

Family	Features	Behaviour
F1	18	constant across the six datasets
F3	14	constant across the six datasets
F4	22	constant across the six datasets
2D image	12	constant across the six datasets
Tabular	5–30	schema-dependent

This constancy is the principal quantitative evidence for the data-contract claim advanced in Section III-D. Dimensionality is a function of the family, not of the individual dataset: six electrocardiographic, encephalographic and polysomnographic collections differing in acquisition hardware, sampling rate, channel count and recording duration all converge to eighteen descriptors, because all of them are normalized into the same contract before extraction begins. Had modality-specific behaviour leaked past the reader layer, dimensionality would vary with the source rather than with the family.

The tabular family is the deliberate exception, and its variability is informative rather than anomalous. Tabular descriptors are the native clinical variables recovered by schema inference, so dimensionality necessarily follows the source schema. Across the six tabular datasets the observed values span five to thirty descriptors, with five of the six datasets yielding sixteen or fewer and a single dataset accounting for the upper end of the range. Tabular input is thus the only family in which the representation is determined by the dataset rather than by the pipeline — which is the expected consequence of the fact that no extraction is performed, only schema recovery.

D. Descriptors Effectively Extracted

Table VII lists the descriptor groups produced for each family, and Table VIII details the composition of the eighteen descriptors emitted for family F1, which is the family combining the largest number of analytical domains.

Table VII
DESCRIPTOR GROUPS PRODUCED PER FAMILY

Family	Descriptors
F1	heart rate, RMSSD, SDNN, pNN50, Welch band power, time-domain statistics
F3	Hounsfield range, high-density fraction, mean gradient, uniformity, GLCM descriptors
F4	lesion volume (mm ³), 3D sphericity, GLCM on three orthogonal planes
Tabular	native clinical variables of the source dataset
2D image	GLCM entropy, GLCM contrast, solidity, circularity

Table VIII
ANALYTICAL DOMAINS COMPOSING THE F1 DESCRIPTOR SET

Domain	Descriptors
Time	RMS, mean, standard deviation, skewness, kurtosis, peak-to-peak and maximum amplitude, SNR (dB)
Frequency	total power, spectral centroid, dominant and median frequency, power in four fixed bands
EEG-specific	relative power in the delta, theta, alpha, beta and gamma bands
ECG-specific	R-peak count, mean heart rate, mean and standard deviation of RR intervals, RMSSD, SDNN, pNN50
EMG-specific	envelope RMS
Spirometry	FVC, FEV1, FEV1/FVC, PEF

Two properties of the descriptor sets are worth noting. First, the sets are layered rather than disjoint: every F1 recording receives the time-domain and spectral basis, and modality-specific descriptors are added on top when the corresponding structure is present in the signal. Second, the imaging families share a common core of texture, morphology and intensity descriptors, to which family F3 adds descriptors that require an acquisition header — notably those expressed in Hounsfield units — and family F4 adds descriptors that require voxel spacing, such as lesion volume in physical units. Descriptor availability is therefore governed by metadata availability, a point returned to below.

E. Discussion

1) *Flexibility versus specialization*: A single pipeline consumed thirty heterogeneous datasets spanning five families without per-dataset reconfiguration, and emitted a well-formed representation for each. The price of that generality is visible in the same result that demonstrates it: each family receives a fixed descriptor budget rather than a set tuned to the individual dataset. A pipeline specialized to mammography would almost certainly extract descriptors better adapted to mammography than the fourteen produced here for every F3 input. The proposition this work advances is not that generality dominates specialization, but that a general ingestion stage is achievable at a characterizable cost, and that the cost is a fixed rather than a per-dataset one.

2) *What the constancy result establishes, and what it does not*: The constant dimensionality within each signal-bearing family establishes that the abstraction boundary holds: no modality-specific behaviour leaked past the reader layer into extraction. It does not, by itself, establish that the descriptors are clinically informative. Those are separate claims requiring separate evidence, and the second is deliberately deferred to the companion work. Reporting the first without the second is a scoping decision rather than an omission.

3) *Detection by structure as a design trade-off*: The two divergent cases expose the principal trade-off of the detection mechanism. Detection by physical structure is deterministic, fast, auditable and requires no content inspection, but it cannot

recover semantics that the encoding does not express. A spirometry study exported as a table of sampled values is, structurally, a table. Three mitigations are available and are not mutually exclusive: content-based inspection of column names and value distributions to identify signal-like tables; an explicit user-supplied family override at ingestion time; and extending the validation layer, which already performs a real header read, to attempt a signal interpretation of tabular input before accepting the tabular assignment. The third is the most consistent with the existing architecture, since it strengthens a stage that is already present rather than adding one.

4) *Metadata availability as a constraint on representation*: Descriptor availability tracks metadata availability. Family F3 supports intensity descriptors expressed in Hounsfield units because the DICOM header supplies the windowing parameters; family F4 supports lesion volume in cubic millimetres because voxel spacing is recoverable. Neither is available for a plain raster image of the same anatomy. This is a property of the input rather than of the pipeline, but it bounds what any modality-agnostic system can promise: normalization can unify structure, and cannot manufacture absent acquisition context.

5) *Threats to validity*: Four limitations qualify these results. The benchmark comprises six datasets per family, which is sufficient to demonstrate that the mechanism operates across families but not to characterize its behaviour within a family in depth. The expected family of each dataset was known in advance, so the agreement reported in Table V measures the detector against curator intent rather than against an independent ground truth. Ingestion time and peak memory consumption were not instrumented in this evaluation, so no claim is made regarding computational cost, particularly for volumetric input, where it is expected to be highest. Finally, the number of samples effectively processed per dataset was not recorded, so the results characterize the pipeline’s behaviour per dataset rather than per sample.

6) *Relation to the companion work*: The feature vectors characterized in this section are the input to the modeling stage evaluated in the companion paper. Whether the eighteen F1 descriptors, the fourteen F3 descriptors or the twenty-two F4 descriptors support accurate classification is a question that this paper poses and does not answer, by construction.

V. CONCLUSION

This paper described the ingestion and feature-engineering stage of a modality-agnostic biomedical pipeline, and evaluated it over thirty real benchmark datasets spanning temporal physiological signals, two-dimensional DICOM studies, volumetric acquisitions, tabular clinical records and plain two-dimensional imaging. The pipeline classifies an arbitrary input into one of nine structural modes, validates it against a family schema by attempting a real header read, normalizes it into a single data contract, applies preprocessing selected from measured properties of the dataset, and emits a multi-domain feature vector. All thirty inputs were processed to completion

without unhandled exceptions, and twenty-eight were assigned to a mode of the family they represent.

A. Contributions

The specific objectives stated in Section I-B are addressed as follows.

The detection mechanism covering nine structural modes, coupled with schema validation, was formalized in Sections III-B and III-C and evaluated in Table V. The unified data contract was defined in Section III-D, and served all five families without modification; its principal empirical support is the constancy of feature dimensionality within every signal-bearing family reported in Table VI, which indicates that no modality-specific behaviour leaked past the reader layer. The adaptive preprocessing stage was specified in Section III-F, where each operation is triggered by a measured property of the dataset and accompanied by a justification recording the value that triggered it. The multi-domain descriptor set was characterized in Section III-G and detailed per family in Tables VII and VIII. The empirical evaluation over thirty datasets constitutes Section IV.

The broader contribution is architectural. By treating the structure of the input as something to be inferred rather than declared, and by converting every inferred family into a single representation before extraction begins, modality-specific knowledge is confined to two layers — the detection rules and the readers — and does not reach orchestration, preprocessing or consolidation. Supporting an additional family therefore requires one reader and one extractor, rather than a revision of the pipeline.

B. Limitations

Four limitations qualify the results, and the first is the most consequential.

Detection operates on the physical organization of the data and cannot recover semantics that the encoding does not express. Two of the thirty datasets — an electromyographic recording and a spirometric study, both distributed as delimited text — were assigned to the tabular mode. That assignment is correct with respect to the encoding observed and incorrect with respect to the clinical meaning of the recordings, and it propagates into the representation, since a signal routed to the tabular mode is described by its column schema rather than by spectral and variability descriptors.

Label recognition depends on a fixed vocabulary of directory names, so a dataset adopting an unanticipated naming convention will not be recognized as labeled even when a class partition exists. The audio and video families are outside the scope of the present version of the system. Finally, ingestion time and peak memory consumption were not instrumented, and the number of samples processed per dataset was not recorded, so no claim is made regarding computational cost — a gap that is most relevant for volumetric input, where cost is expected to be highest.

C. Future Work

Three directions follow directly from the limitations above. The first is to extend the validation layer, which already performs a real header read, so that it attempts a signal interpretation of tabular input before accepting the tabular assignment; this would address the divergent cases without introducing a new architectural stage. The second is to add a learned-representation branch alongside the handcrafted descriptors, using the existing contract as its input, which would allow the two extraction strategies to be compared under identical ingestion conditions. The third is to instrument the pipeline for ingestion time and memory consumption, so that the cost of generality can be quantified rather than merely acknowledged.

Finally, the feature vectors characterized here are consumed by the modeling and reporting stage evaluated in a companion work. Whether the descriptor sets reported in this paper support accurate and well-calibrated clinical classification is a question posed by the present work and answered there.